160 KVA Ashok Leyland - Generator Specifications.

Bore x Stroke (mm)	104 x 113 mm
Rating	160 kVA
Engine Make	Ashok Leyland
Model Number	SG160AL
Noise Level	90 dB
Cooling System	Liquid Cooling
Frequency	50/60 Hz
Battery Capacity	12V 90 Ah
Dimension	3900 x 1350 x 2070 mm
Fuel Tank Capacity	650 Liter
Weight	2575 kg
Electric Power	128 kWe
Rated Current @ 380V	243 A
Engine Model	AL6DTIDG4
Aspiration	TA
Engine Rated kW	138
No. of Cylinders	6
Displacement	5.76 Liter
Coolant Capacity Approx.	20 Liter
Lube Oil Capacity Approx.	17.5 Liter
Lube Oil Specification	SAE15W 40 CI4
Oil Change Period	300 Hours
IP Rating	IP33

Key Features:

- Reliable, Robust & Rugged engines designed to deliver continuously in arduous environment
- Better Block Loading Capability
- Compact engine: Engineered for Optimized Power Solutions
- Governor: Mechanical
- Low Oil & Fuel consumption
- Continuous duty Power rating
- Minimal vibrations and lower noise levels
- Easy Serviceability & Repairability
- Alternator: STAMFORD Brushless, Single Bearing, Class H Insulated, Voltage Regulation: ± 0.5%, 0.8PF Lag

GENERAL DESCRIPTION	UoM	160kVA
New Name after Emission Certification	Engine Model	**AL6DTIDG4**
No of Cylinders	Qty	6
Gross Rated Power	kW/hp	147 / 200
Overload Power (110 %)	kW/hp	161.7 /220
RPM		1800
Typical Alternator Efficiency	% age	92.8
Working Principle	2/4 Stroke	4
BORE / STROKE	mm / mm	104 x 113
Displacement	cc	5760
Compression Ratio	mm	16.5 ± 0.5:1
Unit Dry Wt of Bare Engine Kg	kg	380
Flywheel Housing	SAE	SAE 2
Direction of Rotation from Flywheel End		Counter Clockwise
Firing Order	Seq	1-4-2-6-3-5
Mechanical efficiency	% age	94.0%
Thermal efficiency	% age	40%
Reference Standard	REF	ISO 8178
Governing system		Elecronic
Type of Governor		
Class Of Governor		A0
Engine Speed Droop (Governing)	% age	0%
Heat Rejection to atmosphere	% age	4-7%
Battery Volts	Volts	24V
Battery Capacity	A.H.	105 AH
No. of Starter Motors provided	Qty.	1
Rating Of starter motor	kW	2.8
Battery Charging alternator Capacity amps	amp	35
Intake Manifold Temperature (Max)	deg.C	48 ± 2
Intake Manifold Pressure	kPa	100-180
Intake Air Flow	m3/min	10.25
Max. Allowable restriction in Air Element	inch	14
Air Cleaner efficiency for dirt removal	% age	99.95%
Engine Cooling System Coolant Capacity	Lit.	23
Cooling Heat Rejection, kW	kW	137.7
Max Cooling System Pressure	PSI	12.8
Coolant Outlet Temp (Max)	deg. C	95
Coolant Inlet Temperature	deg. C	75
Start to Open temp for Thermostat	deg. C	82
Fully Open temp for Thermostat	deg. C	95
Coolant topping-up / draining frequency		Every 2 Years or 1000 hrs.
Fan Diameter	inch	28"

Fan Power	kW	7
Manifold Type		Dry Type
Max. Wt of Exhaust System on Engine	kg	24
Dia of Exhaust Outlet (inside)	mm	72
No of Exhaust outlet	Nos.	1
Max allowable System back pressure kPa	kPa	7
Heat rejection to exhaust	kW	94.3
Heat rejection to coolant	kW	137.7
Heat rejection to atmosphere	%	~4to7%
Stack Temp	deg. C	500
SFC at 100%	gms/bhp/Hr	157.18
SFC at 75%	gms/bhp/Hr	161.73
SFC at 50%	gms/bhp/Hr	170.13
No of Holes	no	4
Dia of Hole	mm	11
PCD	mm	125
Fuel Pump		Rotary
Fuel Transfer pump type		Plunger
Filter Type		Spin On
No of filters		1
Refill volume with Lube Filter change	lit	18
Lub Oil system capacity (with filters)	lit	18
Lub Oil temperature (max)	deg. C	120
Fuel Shut-Off solenoid type & Travel		Electrical
Low Lub Oil Pressure Shut-Off	psi	7.5
High Water Tempo Shut-Off	deg.C	105 ± 3
Crank termination signal	r.p.m / Oil. Press	Oil Pressure
Lub Oil Pressure Guage		Digital
Water Temp Guage		Digital
Hourmeter cum r.p.m meter		Digital
Ammeter		Digital
Flywheel Housing Coupling Size	SAE	SAE 2
Flywheel Coupling Size	SAE	SAE 11.5
Flywheel Diameter	mm	436
Flywheel Width	mm	87
Flywheel No of teeth	no.	136
Flywheel Weight	kg	40.1
Noise on Full Load at 1 m	dba	98
Engine Warm-Up time before putting load	Sec	60

Engine Cooling time before Engine Shut-Down	Sec	120
Sudden 100% decrease in Load. =<10%	%	NA
Sudden 100% increase in Load. =<10%	%	NA
Recovery for 100% decrease inload =<8 sec	Sec	NA
Recovery for 100% increase inload =<8 sec	Sec	NA